

Where injected work goes in a hard-disk box: a proof ladder for an energy-transfer experiment

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Abstract—A piston compresses a hard-disk gas in a closed box. We ask what fraction of the injected work can still be captured as ordered energy, and we refuse to answer until each simpler statement below it is closed: the ledger, the quasi-static baseline, the finite-rate excess, the spring, the divider, the chain. Levels 0–2 are closed here. The headline of Level 2 is that a closed compartment has *no* linear friction: the force autocorrelation of a held piston integrates to zero at the sound traversal time, and the finite-rate cost is second order, $\langle W \rangle - W_{\text{qs}} = Au^2$, of which roughly half is energy the protocol itself launches.

Index Terms—hard-disk fluid, energy transfer, event-driven molecular dynamics, dissipation, adiabatic piston

I. THE BOX AND THE LADDER

Geometry A: two compartments of $N_s = 50$ disks either side of an immovable divider, a piston on the right. $k_B T = m = \sigma = 1$, $H = 10$, compartment 38.75σ , gas compression $\Delta x = 3.93\sigma$.

II. LEVEL 0 — THE LEDGER CLOSES

$R_E = W - [\Delta K E_{\text{gas}} + \Delta K E_{\text{walls}} + \Delta E_{\text{spring}} + Q]$ and the matching momentum residual, from the per-event impulse log. **Measured:** $|R_E|/W \leq 8 \times 10^{-12}$, $|R_p|/\sum |\Delta p| \leq 1 \times 10^{-13}$ over all 75 pilot trajectories. **Passed.**

III. LEVEL 1 — THE QUASI-STATIC BASELINE

$W_{\text{qs}} = N_s k_B T (T_f/T_i - 1)$ with $\ln(T_f/T_i) = \int Z d \ln \eta$ along the path, Z measured in the finite box rather than taken from the bulk EOS. **Measured:** $W_{\text{qs}}^{\text{finite}} = 7.4685 \pm 0.0131 k_B T$; extrapolating the measured work to $u \rightarrow 0$ with the (quadratic) form gives $W(0) - W_{\text{qs}} = +0.0043 \pm 0.0158 k_B T$, 0.3σ . **Passed**, and on the physical side.

IV. LEVEL 2 — THE FINITE-RATE EXCESS

A. Slow end

$$\zeta(\lambda) = \beta \int_0^\infty \langle \delta F(0) \delta F(t) \rangle dt, \quad W_{\text{diss}} \simeq \zeta u \Delta x. \quad (1)$$

Measured: the integral is not Markovian. It falls from 2.55 at zero lag — the Enskog collisional friction — and is cancelled by the returning sound wave at $L/c_s = 19.8\sigma$, confirmed at the compressed end of the path where the crossing moves to 18.0σ against a predicted 15.9. Its zero-frequency limit is the hydrodynamic viscous friction $\zeta_{\text{hyd}} = L_y \Gamma / X_p = 0.081 \pm 0.010$,

which predicts a linear term of $0.32 \pm 0.04 u$ — below the resolution of 760 trajectories, and consistent with the measured slope -0.50 ± 0.67 for $u \leq 0.05$. Γ is taken from Paper 1’s divider linewidth, not assumed.

What is there instead is quadratic: $A_{\text{step}} = 25.9 \pm 4.3$ against $A_{\text{ramp}} = 11.3 \pm 2.5$ when the piston is ramped over $2L/c_s$ instead of stepped (3.0σ apart). Roughly 14 of the 25 is energy the step protocol launches; what a gentle push leaves is consistent with the steady-flow value $N_s m / 6 = 8.3$.

B. Fast end

$\langle W \rangle \simeq 2N_s m (\Delta x / L) u^2$, no free parameter. **Measured:** 0.986 of prediction at $u = 10$, $\Delta x = 7.08\sigma$. The apparent 8% excess at short travel is the contact layer: the strip a face can strike (width Δx starting at face $-r$) holds $1.13\times$ the uniform-density count at $\Delta x = 0.75\sigma$. Using the measured count the ratio becomes 0.96–1.03 with no trend.

V. LEVEL 3 — CAN A SPRING CAPTURE IT?

Open; v4 pending. Geometry C: one gas of 100 disks drives a spring-loaded wall through a 10% compression ($\Delta x = 7.96\sigma$, η 0.100 $\rightarrow$ 0.111).

The apparatus is a pre-loaded spring. It holds the standing gas force $F = N k_B T Z / L$, so the wall sits $d = F/k$ from the anchor and $\Delta E = F s + \frac{1}{2} k s^2$ is *first* order in the wall displacement — the $F s$ term is work lifting a weight. Limits: $k \rightarrow 0$ is a pushrod ($\epsilon \rightarrow 1$ trivially, nothing taken from the gas), $k \rightarrow \infty$ a rigid wall ($\epsilon \rightarrow 0$).

The gas spring is adiabatic, and it is Paper 1’s c_s . With $U = N k_B T$ in 2D, $dU = -P dV$ gives $d \ln T / d \ln L = -Z$ and $k_{\text{gas}} = (N k_B T / L^2) c_s^2$ — which is also the heavy-divider limit of Román’s $\cot K = \alpha K$, so the two papers share one line of physics. It is $2.01\times$ the isothermal stiffness here, and the adiabatic reproduces Level 1’s measured $T_f/T_i = 1.149$ to 0.5%.

Status. The ledger closes (median 4.6×10^{-10} , 593/600 below 10^{-8}) and the coherent peak stands $6.8\text{--}31\sigma$ above the pedestal. A parameter-free one-degree-of-freedom model $M \ddot{s} = F_{\text{ad}}(L - x_p + s) - F_{\text{ad}}(L) - k s$ reproduces the *settled* wall displacement to 8–10% but under-predicts the *peak* by 19–57%. The residual is therefore entirely in the transient, and it is the acoustic pulse: the model is undamped, so it already gives the largest overshoot a linear system can. Closing it needs the wall’s radiation reaction in the drive.

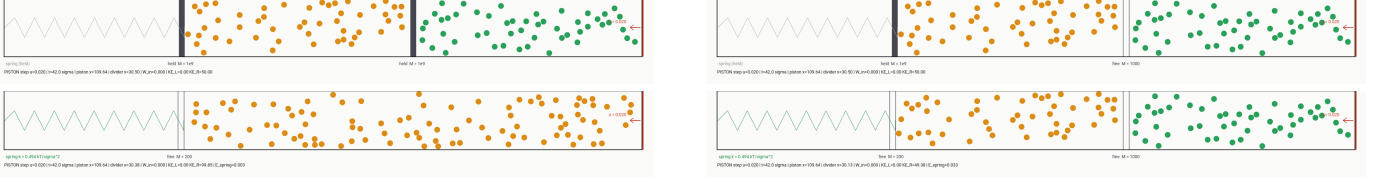


Fig. 1. The master box, one apparatus with elements switched on. A (top left) both walls held, as Levels 0–2 were measured; B (top right) divider free; C (bottom left) divider removed, one gas driving the spring-loaded wall; D (bottom right) both free. Held walls are drawn solid and labelled by mass, free walls hollow; the spring is grey when held and green when live.

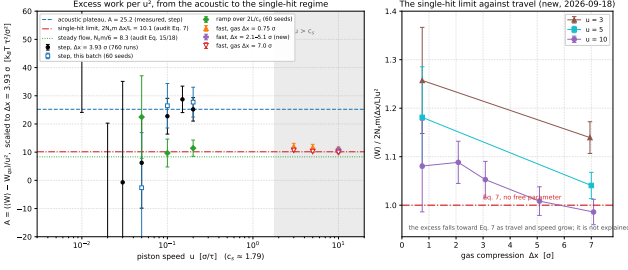


Fig. 2. $A(u) = (\langle W \rangle - W_{qs})/u^2$ from the acoustic to the single-hit regime, every point scaled to a common travel, with the travel scan at right.

VI. LEVEL 4 — TRANSMISSION THROUGH A FREE DIVIDER

Not started. Fast pressure channel against slow heat channel; $\mathcal{T}(t) = \Delta KE_{\text{gas}2}(t)/W_{\text{in}}$, with $\mathcal{T}_{\text{slow}} \rightarrow 1/2$ for symmetric gases.

VII. LEVEL 5 — THE CHAIN

Not started. $\epsilon = E_{\text{spring,max}}/W_{\text{in}}$ against piston speed, divider mass and the number of walls; the prediction is the *ordering*, geometric intermediate masses beating badly matched ones.

VIII. LEVEL 6 — INFORMATION

Not started, and gated on a canonical seeder. $\beta \langle W_{\text{diss}} \rangle \geq I_{\text{mem}} - I_{\text{pred}}$. Note that the Jarzynski benchmark must be the finite-box ΔF , exactly as Level 1 needed the finite-box W_{qs} .

REPRODUCIBILITY

Raw trajectories are not in the repository; each cell carries `summary.csv` and `00_COMMAND.md`, and Level 2 an 85kB `stop_digest.csv` from which the entire analysis reproduces. Batch scripts are in `../experiments/run_scripts/`.